

Project Initialization and Planning Phase

Date	28 September 2026
Team ID	SWTID-2026-7718
Project Title	AI Study Buddy API
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report.

AI Study Buddy is a web-based learning assistant that uses Artificial Intelligence to provide students with instant answers, simple explanations, and study support.

Project Overview	
Objective	To provide quick and easy academic assistance and help students understand difficult concepts.
Scope	The system supports question answering, topic explanations, material summarization, flashcards, quizzes, and study planning.
Problem Statement	
Description	Students often face difficulty understanding complex topics and finding quick, reliable study assistance.
Impact	The problem can cause confusion, stress, and increased time spent searching for study materials.
Proposed Solution	
Approach	Develop a web-based AI Study Buddy that accepts student questions and uses AI/NLP to generate relevant and easy-to-understand responses.
Key Features	☐ AI-based question answering ☐ Simple explanation of difficult topics

	☐ Study material summarization ☐ Flashcards and quizzes ☐ Personalized study plans
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing resources	Computer/Laptop
Memory	RAM	8 GB
Storage	Project files and database	256 GB or above
Software		
Frameworks	Used to build the backend and handle requests between the user interface and AI system.	Node.js, Express.js
Libraries	Provides AI and NLP functionality for processing student questions and generating responses.	AI/NLP libraries
Development Environment	Used to write, test, and manage the project code.	Visual Studio Code
Data		
Data	Contains student queries, user information, and study-related content.	MongoDB